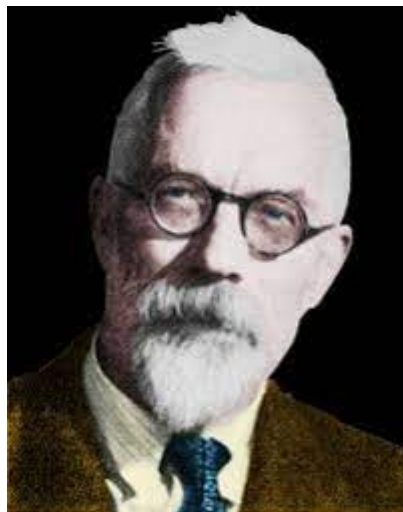


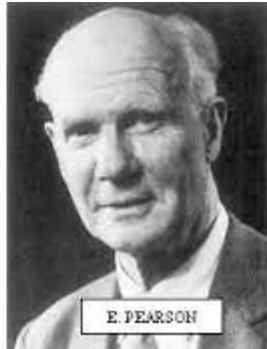
Fisher Tradition & Probability

"Fisher" tradition



- Set up a statistical null hypothesis (note that null does NOT mean "nil")
- Report the exact level of significance
- Do not use a "conventional" level, do not talk about accepting/rejecting hypotheses, do not pass GO, and do not collect \$200
- Use this procedure *only* if you know very little about the problem at hand

Neyman-Pearson



- Set up 2 hypotheses, and design a study based on the "rejection region" for each hypothesis
- If data is within the rejection range for H_1 , accept H_2 . Otherwise, accept H_1 . Note that accepting it doesn't mean you *believe* it...just that you act as though it was so
- Utility is limited to situations where there is a clear difference in hypotheses, when you can make a rational decision about when to accept vs. when to reject H_1 and H_2

So who won?

The two ideas melded together somehow into something that neither camp would be too excited by: 1) Set up a null hypothesis, where null almost always means "chance" 2) Make a yes-no decision about that hypothesis 3) Repeat

We want to know:

- what is the probability that we would get the values evidenced (*or those more extreme*) given our null hypothesis
- 🙌 $P(Data + | H_0)$ aka a **p-value**
- assumes, among other things, that the **null hypothesis is exactly true**, that you have a **random sample**, and that the scores are **independent**

Probability



Sample Space & Assumptions

Our sample space is the range of possible values for a random variable. 6 Clue characters.

Assumption 1) Sum of all the probabilities of all outcomes needs to equal 1. $P(S) = 1$

Assumption 2) The probability of an event occurring must be between 0 and 1. $0 \leq P(event) \leq 1$

P(Miss Scarlet)



- $P(\text{Miss Scarlet}) = \text{N of events} / \text{sample size}$
- $P(\text{Miss Scarlet}) = 1 \text{ Miss Scarlet} / 6 \text{ characters}$
- $P(\text{Miss Scarlet}) = 1/6$

P(Female)



- $P(\text{Female}) = \text{N of events} / \text{sample size}$
- $P(\text{Female}) = 3 \text{ females} / 6 \text{ characters}$
- $P(\text{Female}) = 3/6 = .5$

Complement

- The probability that the event does *not* occur
- $1 - P(event)$

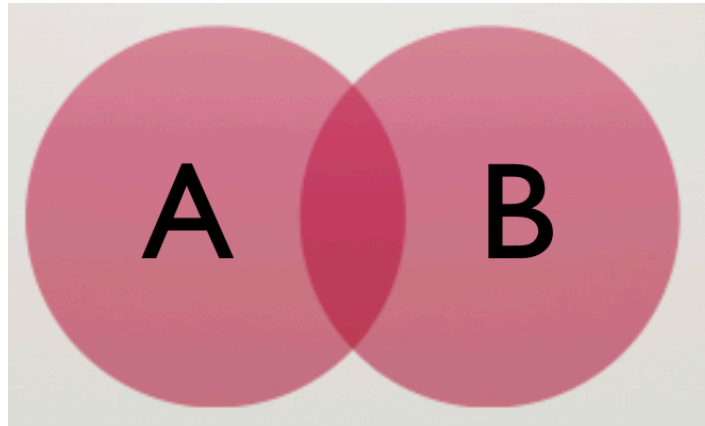
P(NOT Female)



- $P(\text{NOT Female}) = N \text{ of events} / \text{sample size}$
- $P(\text{NOT Female}) = 3 \text{ not females} / 6 \text{ characters}$
- $P(\text{NOT Female}) = 3/6 = .5$

Unions

- The possibility of A **or** B occurring
- All elements that are in one of A or B
- $P(A \cup B)$



- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

P(Female or Holding Something)



P(Female or Holding Something)

P(Female)

- $P(\text{Female}) = 3/6$

P(Holding Something)

- $P(\text{Holding Something}) = 4/6$

P(Female & Holding Something)

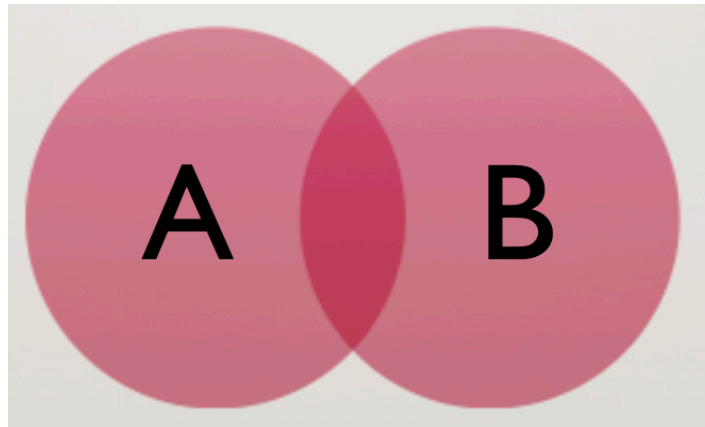
- $P(\text{Female \& Holding Something}) = 2 \text{ females with stuff} / 6 \text{ characters}$
- $P(\text{Female \& Holding Something}) = 2/6$

P(Female OR Holding Something)

- $P(\text{Female or Holding Something}) = 3/6 + 4/6 - 2/6$
- $P(\text{Female or Holding Something}) = \mathbf{5/6}$

Intersection

- The probability of A *and* B occurring
- $P(A \cap B) = P(A) \times P(B|A)$



P(Baker & Female)

- $P(\text{Baker}) = 1/6$
- $P(\text{Female} \text{ GIVEN there is a baker}) = 1 \text{ baker that's female} = 1$
- $P(\text{Baker \& Female}) = 1/6 * 1 = 1/6$

Intersection

- The probability of A *and* B occurring
- $P(A \cap B) = P(A) \times P(B|A)$
- P(Baker & Female) has *dependent events*; the occurrence of Event A changes the probability of Event B
- *Independent* events would be that the occurrence of Event A does NOT impact the occurrence of Event B
- *If independent*, $P(A \cap B) = P(A) \times P(B)$

Independence of observations is one of the criteria for having interpretable p -values!

- 2 games of Clue
- Finding the murderer for game 1 doesn't help you find the murderer for game 2
- $P(\text{Murderer in Game 1} \ \& \ \text{Murderer in Game 2}) =$
 $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$

Permutations

All the ways n objects can be arranged

$$N = 2$$

```
##      01 02
## 1    b  a
## 2    a  b
```

Permutations

All the ways n objects can be arranged

$$N = 3$$

##		01	02	03
##	1	c	b	a
##	2	b	c	a
##	3	c	a	b
##	4	a	c	b
##	5	b	a	c
##	6	a	b	c

Permutations

$$N = 4$$

##		01	02	03	04
## 1		d	c	b	a
## 2		c	d	b	a
## 3		d	b	c	a
## 4		b	d	c	a
## 5		c	b	d	a
## 6		b	c	d	a
## 7		d	c	a	b
## 8		c	d	a	b
## 9		d	a	c	b
## 10		a	d	c	b
## 11		c	a	d	b
## 12		a	c	d	b
## 13		d	b	a	c
## 14		b	d	a	c
## 15		d	a	b	c
## 16		a	d	b	c
## 17		b	a	d	c
## 18		a	b	d	c
## 19		c	b	a	d
## 20		b	c	a	d
## 21		c	a	b	d
## 22		a	c	b	d
## 23		b	a	c	d

Permutations

The number of permutations for n objects is:

$$n! = n(n - 1)(n - 2)\dots$$

BUT, if you're looking for the number of r choices from n :

$$\frac{n!}{(n - r)!}$$

There are 6 Clue characters, but we are now 100% sure that 2 of them are the culprits. How many possibilities are there?

- $n = 6, r = 2$
- $\frac{6!}{(6-2)!} = \frac{6!}{4!}$
- $\frac{720}{24} = 30$

Combinations

1 Permutation = Miss Scarlet & Professor Plum. Another permutation is Professor Plum and Miss Scarlet. What if we don't care who comes first and who comes second? The pairing of Plum/Scarlet should be good enough!

When order doesn't matter, we count the number of **combinations**

$$\frac{n!}{(n-r)!r!}$$

- $n = 6, r = 2$
- $\frac{6!}{(6-2)!2!} = \frac{6!}{4!2!}$
- $\frac{720}{24*2} = \frac{720}{48} = 15$
- We have a 1/15 chance of getting the correct 2 culprits out of 6 characters

Combinations on Combinations

We split up the class into 2 different games of Clue. In the first game, we assume 2 culprits. In the second game, we assume 3 culprits. The intersection of these combinations is the probability that we get both the 2 culprits correct in game 1 and the 3 culprits correct in game 2.

Game 1

- $n = 6, r = 2$
- $\frac{6!}{(6-2)!2!} = \frac{6!}{4!2!}$
- $\frac{720}{24*2} = \frac{720}{48} = 15$
- $1/15$

Game 2

- $n = 6, r = 3$
- $\frac{6!}{(6-3)!3!} = \frac{6!}{3!3!}$
- $\frac{720}{6*6} = \frac{720}{36} = 20$
- $1/20$

$$\frac{1}{15} \times \frac{1}{20} = \frac{1}{300}$$

Bernoulli Trials

- Exactly 2 possibilities, **success** or **failure**
- Flipping a coin. Heads = Success, Tails = Failure
- Let's say we flip a coin twice. Our possibilities are: TT, TH, HT, HH...

Bernoulli Trials

How many ways are there to get 0 Heads out of 2 coin flips?

- $\frac{2!}{(2-0)!0!} = 1$

How many ways are there to get 1 Heads out of 2 coin flips?

- $\frac{2!}{(2-1)!1!} = 2$

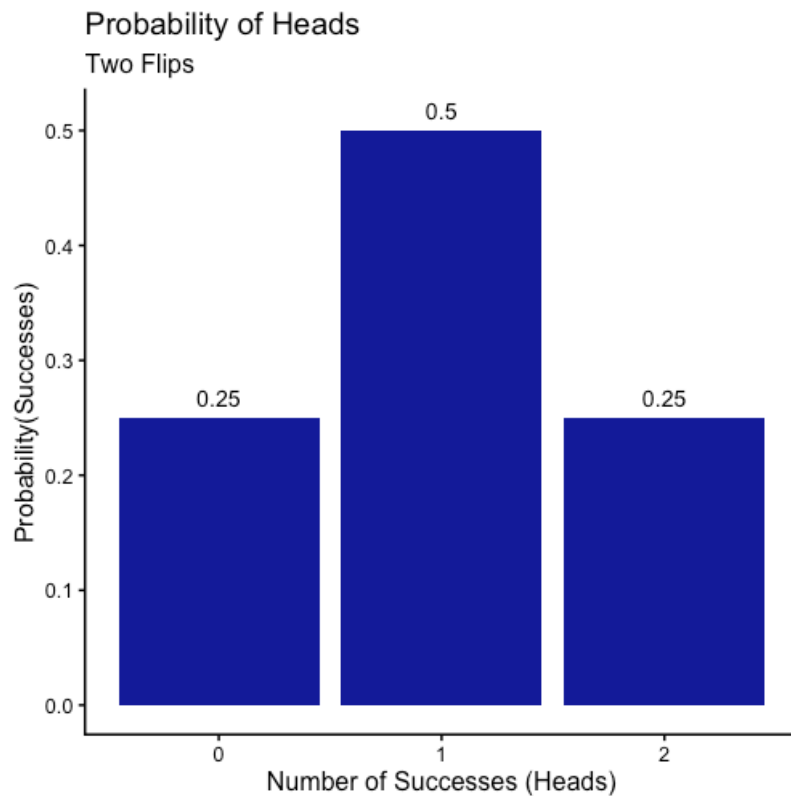
How many ways are there to get 2 Heads out of 2 coin flips?

- $\frac{2!}{(2-2)!2!} = 1$

Bernoulli Trials

How many outcomes are there total?

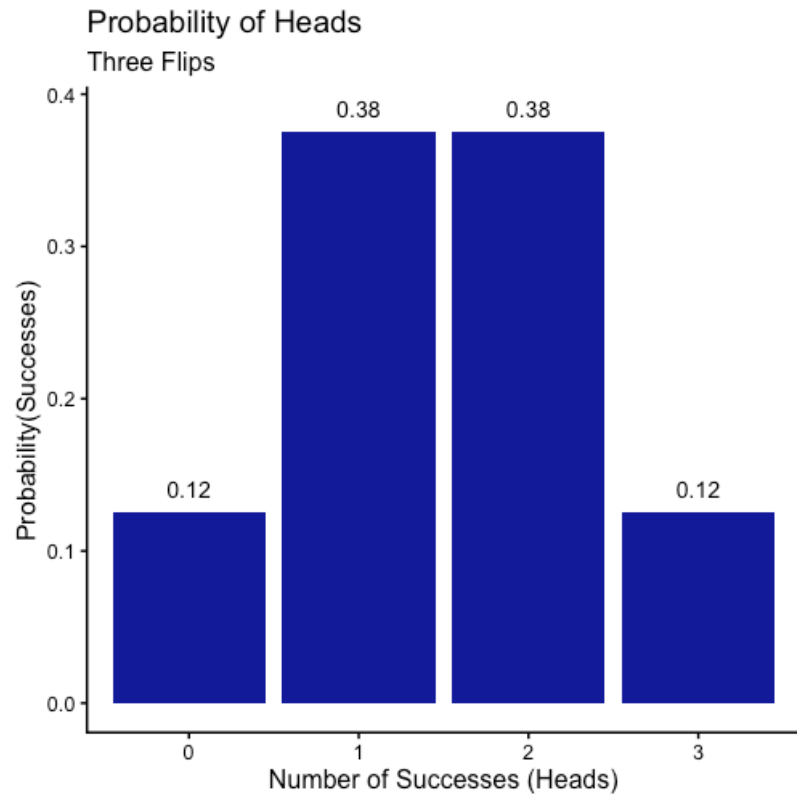
- $1 + 2 + 1 = 4$
- Probabilities = $1/4$, $2/4$, $1/4$



Bernoulli Trials

Three coin flips?

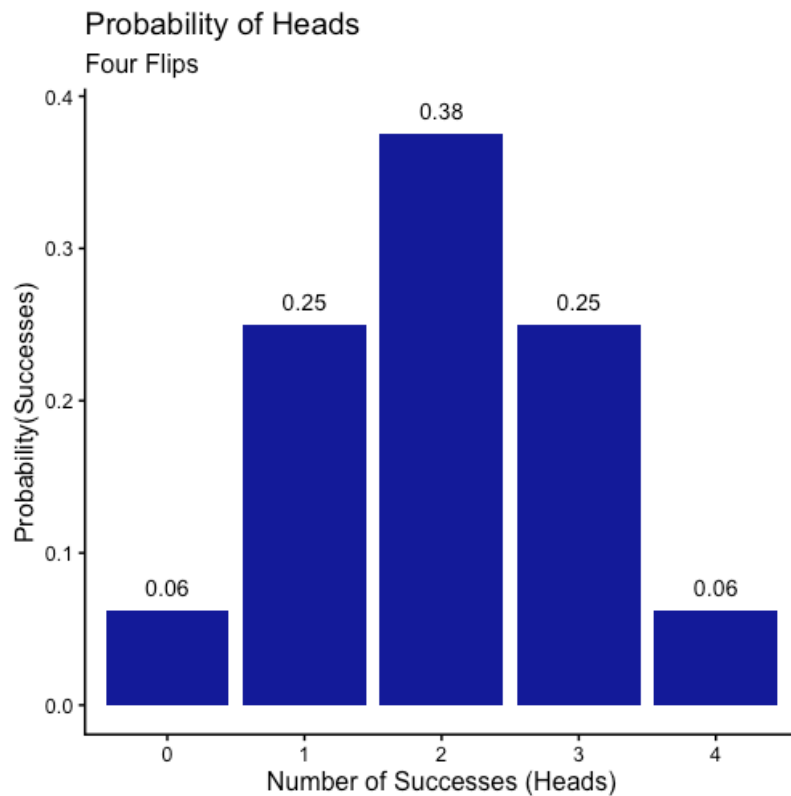
- TTT, TTH, THT, THH, HTT, HTH, HHT, HHH



Bernoulli Trials

Four coin flips?

Probability Mass Function



Counting Rule: Combinations

- Previous we used r for choices. Let's be more specific. Let k be the number of *successes*
- Total number of ways of getting k successes out of n trials, irrespective of order:

$$\frac{n!}{k!(n-k)!} = \binom{n}{k}$$

- Called **binomial coefficient**
- Go back a few slides...we just did this...a lot

What is the probability of getting 3 heads out of 5 coin flips?

Tails Heads Heads Tails Heads

This can be represented in binomial form. First, we need to choose which value represents a "success". Here, we'll use **Heads**.

NotHeads Heads Heads NotHeads Heads

The probability of that particular sequence is:

$$P(\text{NotHeads})P(\text{Heads})P(\text{Heads})P(\text{NotHeads})P(\text{Heads})$$

$$P(\text{Heads})^3 P(\text{NotHeads})^2 = \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = 0.03125$$

What is the probability of getting 3 heads out of 5 coin flips?

But a specific sequence of independent outcomes is just one way we could have X successful trials out of N

- We need to know **how many possible ways** we could get X successes in N trials
 - HHHTT, HTHTH, TTHHH etc...

Remaining part of the equation is the combination rule for probability theory, $\binom{N}{k}$, it tells us how many different ways this can happen

H	H	H	T	T
H	T	H	H	T
H	T	H	T	H
H	H	T	H	T
H	H	T	T	H

T	T	H	H	H
T	H	T	H	H
T	H	H	T	H
T	H	H	H	T
H	T	T	H	H

$$\frac{n!}{k!(n-k)!} = \frac{5!}{3!(5-3)!}$$

$$\frac{5!}{3!2!} = \frac{120}{12} = 10$$

What is the probability of getting 3 heads out of 5 coin flips?

$$P(X = \text{a head, three times} | p=.5, n = 5)$$

$$= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

$$= \frac{5!}{3!(5-3)!} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2$$

$$= (10)(.03125)$$

$$= .3125$$

Or in R

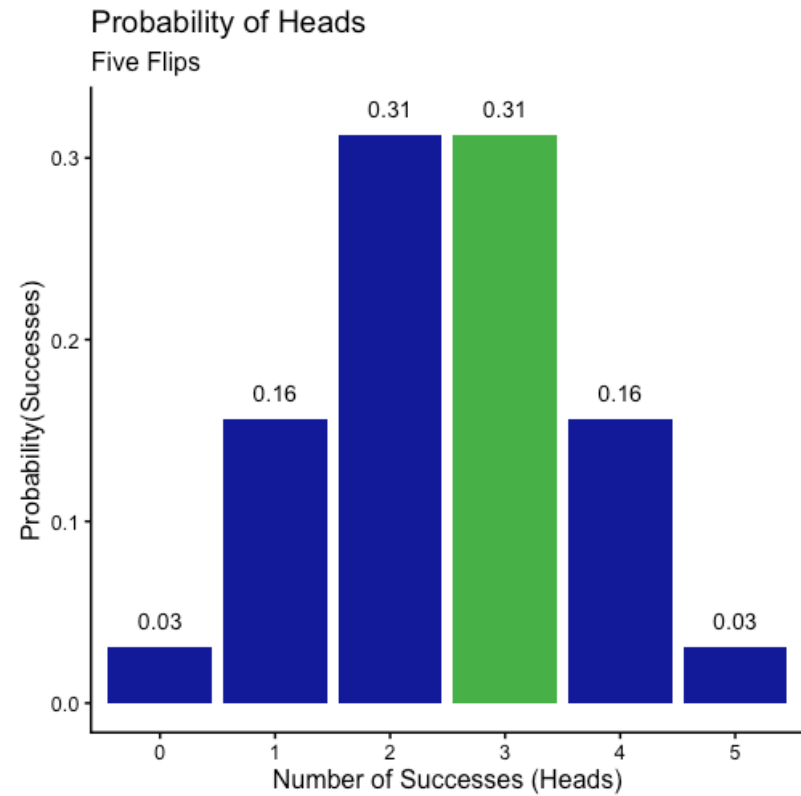
```
dbinom(x = 3, size = 5, prob = .5)
```

```
## [1] 0.3125
```



```
data.frame(heads = 0:5, pr
```

##	heads	prob
## 1	0	0.03125
## 2	1	0.15625
## 3	2	0.31250
## 4	3	0.31250
## 5	4	0.15625
## 6	5	0.03125



Next time...

Continuing with the Binomial Distribution